

Personal Development

Semester 6: Advanced Technology

Fontys University of Applied Science

Eindhoven, The Netherlands

Date	:	14.12.2023
Version	:	0.5
Status	:	Progress
Author	:	Casey Stijlaart

Table of Contents

1	Persona	3
2	Context	4
3	Project	5
3.1	Top 3 Projects	5
3.2	Assigned Research Project: (IoT) measurement setup for observing lab set-ups	5
3.3	Personal Project.....	5
4	Learning outcome table with proof.....	6
4.1	Products with their description	6
4.1.1	Project Management - Git	6
4.1.2	Project Plan.....	6
4.1.3	Database Research	6
4.1.4	Algorithm Research	6
4.1.5	Component Design	6
4.1.6	Test Application Database	6
4.1.7	System Design Document.....	6
4.1.8	Test Plan	6
4.2	Learning Outcome Table	7
5	Sprint retrospective and personal reflection	9
5.1	Sprint 1	9
5.1.1	Retrospective.....	9
5.1.2	Reflection.....	9
5.2	Sprint 2	10
5.2.1	Retrospective.....	10
5.2.2	Reflection.....	10
5.3	Sprint 3	11
5.3.1	Retrospective.....	11
5.3.2	Reflection.....	11
5.4	Sprint 4	12
5.5	Sprint 5	12
6	Evaluation and reflection on the whole semester	12
7	Feedpulse	13

1 Persona

Name: Casey Stijlaart (He/Him)

Age: 23

Personality Type: ISTJ-A

Photograph:



Background:

In addition to my specialization in technology and embedded systems, I have also delved into various aspects of ICT through coursework and practical projects. This well-rounded education has equipped me with an understanding of various programming languages, including C, C++, and Python, as well as expertise in software design. Furthermore, my studies, including my internship, have helped me refine my overall skills in software development and how to adapt to new situations, regarding the project I am working on.

Interests:

Beyond my education, I find comfort and inspiration in a variety of ways, each provided my interests. One of my main interests is thinking about new approaches and how to challenge current challenges within our society. How one can help improve the current situation, or how one can make the life of another individual easier. Looking at more practical things that keep me busy during my time off.

Different things bring me peace, such as being surrounded by nature, gaming, and having a creative outlet.

Skills:

1. **Communication:** I value effective communication, both written and verbal. I have experience collaborating on group projects, presenting ideas, and conveying complex technical concepts in an accessible manner.
2. **Critical Thinking:** My educational journey has enhanced my critical thinking skills, enabling me to analyze problems from multiple perspectives and devise innovative solutions.
3. **Teamwork:** Collaborative projects have taught me the importance of teamwork and cooperation. I'm adept at working in diverse groups and leveraging the strengths of each team member to achieve common goals.

2 Context

One of the contexts that deeply resonates with me is the usage of technology in healthcare. My personal experiences as someone who relies on various aspects of healthcare have made me aware of the importance of improving the current state of healthcare, and I believe that technology plays an important role in achieving this.

My motivation to work in this context is twofold. Firstly, I am driven by the desire to contribute to the betterment of healthcare systems, particularly in enhancing patient outcomes and experiences. For instance, the integration of robotics in surgeries presents an exciting opportunity to decrease surgical risks and improve overall results. This not only captivates my interest but also aligns with my belief that technology can be harnessed to save lives and improve the quality of care. In essence, my personal drive stems from a deep-seated belief in the transformative potential of technology in healthcare. I aspire to actively contribute to this field, improving the lives of individuals and helping shape the future of healthcare through innovation and responsible technological integration.

3 Project

3.1 Top 3 Projects

1. (IoT) measurement setup for observing lab set-ups

This project holds significant appeal for me due to its potential to enhance scientific research and experimentation. The integration of IoT technologies in lab environments not only enables real-time data collection but also can improve efficiency.

2. Autonomous exploration of a building

The concept of autonomous systems navigating and interacting with their environment is at the forefront of technological advancement. This project presents an exciting opportunity to delve into how the way of exploration can be improved and made safer for any involved party.

3. Autonomous Guided Vehicle Platform

The prospect of developing AGV platform sparks my interest as it gives the opportunity to challenge the capabilities of an AGV system. How the system can efficiently navigate and interact with diverse environments.

3.2 Assigned Research Project: (IoT) measurement setup for observing lab set-ups

For the group project I got assigned to my first choice, the (IoT) measurements setup for observing lab set-ups. Within this project, the task is to create a prototype for the measurement setup. The measurement setup project focuses on measuring different aspects of a lab setup, storing the retrieved data in a database, and allowing groups to access the specified data. The group, that requested the data, should only be able to retrieve the data they are allowed to see, meaning that not all data is accessible for each group.

Looking at my personal goals for leaning throughout this project, the focus will be on the following aspects :

- Project Management : As I have been always been in the background when working on projects, I would like to step up more. Getting more of a feel for how to handle and manage a group project. This goal will push me to also be more communicative and take a different approach when things aren't going as planned.
- Safety and Security: In prior projects, the focus was never on creating a system that takes security into account. Therefore I want to focus more on this aspect while working on this project, as within this project security is a more important aspect to be covered.

3.3 Personal Project

For the personal project, a system is created that takes environmental data, such as temperature and sound levels, and creates musical output with the help of algorithms and IoT connectivity.

With the help of this project, I want to broaden my horizons with aspects I haven't worked with before, such as using algorithms to create an output based on input. The introduction of new tools will, hopefully, allow me to get more comfortable with approaching and tackling problems I have not dealt with before.

Another personal of learning is to be more research-based, as prior projects usually already had a direction given. Certain rules and requirements were already set, and therefore only specific research

had to be made. With this project, I would like to do more research on the possibilities out there and which is the most suitable.

4 Learning outcome table with proof

4.1 Products with their description

4.1.1 Project Management - Git

Description: A comprehensive project plan for the personal project, including problem analysis, objectives, timeline, and risk management.

Name in Canvas: PP+GP-project-management.pdf , PP+GP-git-pipeline.pdf

4.1.2 Project Plan

Description: A comprehensive project plan for the personal project, including problem analysis, objectives, timeline, and risk management.

Name in Canvas: PP-project-plan.pdf

4.1.3 Database Research

Description: Research on database storage options, data structure techniques, and recommendations for your group project.

Name in Canvas: GP-research-database.pdf

4.1.4 Algorithm Research

Description: Research on various algorithms for generating music based on environmental data.

Name in Canvas: PP-research-algorithms.pdf

4.1.5 Component Design

Description: Component-based design for a distributed system to create musical output from environmental data.

Name in Canvas: PP-design-component.pdf

4.1.6 Test Application Database

Description: A test application for evaluating database functionality, with a focus on personal and group project requirements.

Name in Canvas: PP-Test-App-Database.png , PP-Test-App-Database.png , GP-test-application-automated-connection.mp4 , GP-Application-Database-Access-Control.mp4

4.1.7 System Design Document

Description: Two comprehensive system design documents. In which various system architecture, design, and behavior are discussed.

Name in Canvas: PP-system-design-document.pdf , GP-system-design-document.pdf

4.1.8 Test Plan

Description: Two comprehensive system design documents. In which various system architecture, design, and behavior are discussed.

Name in Canvas: PP-test-plan.pdf

4.1.9 Code Examples

Description: A comprehensive document, which showcases various code implementations.

Name in Canvas: GP-Code-Examples.pdf

4.1.10 Message Protocol

Description: A comprehensive document, which discusses the message structure and message types, that can occur within the IoT data lab group project.

Name in Canvas: GP-Message-Protocol.pdf

4.1.11 Generation of Musical Composition

Description: A test application for evaluating the generation of musical composition, with a focus on the requirements.

Name in Canvas: PP-Output-AlogorithmicComposition-Poor.mp3, PP-Output-AlogorithmicComposition-Good.mp3

4.2 Learning Outcome Table

Learning outcome	Proof	Rating
LO1: Analysis You specify a distributed computer system including timing, resource use and performance, taking into account safety aspects. And you compile an acceptance test plan and an integration test plan	Product: Database Research: GP-research-database.pdf Product: Test Application Database : PP-Test-App-Database.png , PP-Test-App-Database.png Product: System Design Document: PP-system-design-document.pdf Product: System Design Document: GP-system-design-document.pdf Product: Test Plan: PP-test-plan.pdf	Current: Beginning Self:
LO2: Advise Based on your analysis, you provide technical advisory on the (distributed) computer system that is to be realized, including hardware and software components and links.	Product: Algorithm Research: PP-research-algorithms.pdf Product: Component Design: PP-design-component.pdf Product: Database Research: GP-research-database.pdf Product: System Design Document: PP-system-design-document.pdf Product: System Design Document: GP-system-design-document.pdf Product: Message Protocol: GP-Message-Protocol.pdf	Current: Beginning Self:
LO3: Design You design a distributed computer system including determining actuators, sensors, timing, resource usage, and performance.	Product: Database Research: GP-research-database.pdf Product: System Design Document: PP-system-design-document.pdf Product: System Design Document: GP-system-design-document.pdf Product: Message Protocol: GP-Message-Protocol.pdf	Current: Beginning Self:
LO4: Realisation You realize a complete computer system (or parts thereof) including network, hardware, and system software based on your own design. And you compile and carry out an acceptance procedure to validate the implementation.	Product: Test Application Database: PP-Test-App-Database.png, PP-Test-App-Database.png Product: Test Plan: PP-test-plan.pdf Product: Test Application Database: GP-test-application-automated-connection.mp4 Product: Generation of Musical Composition:PP-Output-AlogorithmicComposition-Poor.mp3, PP-Output-AlogorithmicComposition-Good.mp3 Product: Test Application Database: GP-Application-Database-Access-Control.mp4 Product: Test Application Database: GP-test-application-automated-connection.mp4	Current: Beginning Self:

LO5: Manage & Control You set up and make use of a system for version management, release management, teamwork support, and automated testing for hard- and software systems.	Product: Project Management - Git: PP+GP-project-management.pdf Product: Project Management - Git: PP+GP-git-pipeline.pdf Product: Test plan: PP-test-plan.pdf	Current: Proficient Self:
LO6: Future-oriented organisation You explore the organizational context of ICT assignments, make business, sustainable, and ethical considerations, and manage all aspects of the execution of the assignment.	Product: Database Research: GP-research-database.pdf Product: System Design Document: PP-system-design-document.pdf Product: System Design Document: GP-system-design-document.pdf Product: Code Examples: GP-Code-Examples.pdf	Current: Beginning Self:
LO7: Investigative problem solving You critically consider IT assignments from different perspectives, identify problems, find an effective approach and come up with appropriate solutions.	Product: Algorithm Research: PP-research-algorithms.pdf Product: Component Design: PP-design-component.pdf Product: Database Research: GP-research-database.pdf Product: Code Examples: GP-Code-Examples.pdf	Current: Beginning Self:
LO8: Personal Leadership You are entrepreneurial with regard to ICT assignments and personal development, pay attention to your own learning ability, and keep in mind what kind of ICT professional and/or what type of positions you aspire to.	Product: Project Plan: PP-project-plan.pdf Product: Personal Development: personal-development-document.pdf	Current: Proficient Self:
LO9: Targeted Interaction You determine which partners play a role in the ICT assignment, collaborate constructively with them, and communicate appropriately to achieve the desired impact.	Product: Project Plan: PP-project-plan.pdf Product: Project Management - Git:PP+GP-project-management.pdf	Current: Beginning Self:

5 Sprint retrospective and personal reflection

5.1 Sprint 1

While being ill for a long period of time during this sprint, some conclusions can be made regarding the retrospective and reflection.

5.1.1 Retrospective

What went well:

- After some struggles, the team for the group project came back together as a team and became more communicative.

Areas for improvement:

- As I have been sick for a while, I could improve in being more communicative on the situation and what can be expected of me, towards the team and myself.

Action items:

- Improve planning : I want to create a clear board of tasks to help with an easier way of planning what the current goal is and what is next. Ensuring there is always something to do.

5.1.2 Reflection

What went well:

- While there were struggles in the beginning with communication and a clear direction, I took it upon myself to be a guide as there was no clear "lead" in the group itself. This does resonate with my personal goal of project management, as I have been pushed to broaden my horizon on how we could make this project head in a better direction.
- Within the projects I have been more focused on what is out there instead of the things I know and am more comfortable with. Meaning that I have been more focused on researching possibilities instead of going with what I know.

Areas for improvement:

- While I have been focused on taking a more research-based approach, I still tend to be held back due to wanting to do what I know.

Action items :

- Explore the options on how to help improve project management.

5.2 Sprint 2

5.2.1 Retrospective

What went well:

- After some struggles, the team for the group project came back together as a team and became more communicative.

Areas for improvement:

- As I have been sick for a while, I could improve in being more communicative on the situation and what can be expected of me, towards the team and myself.

Action items:

- Improve planning : I want to create a clear board of tasks to help with an easier way of planning what the current goal is and what is next. Ensuring there is always something to do.

5.2.2 Reflection

What went well:

- While there were struggles in the beginning with communication and a clear direction, I took it upon myself to be a guide as there was no clear "lead" in the group itself. This does resonate with my personal goal of project management, as I have been pushed to broaden my horizon on how we could make this project head in a better direction.
- Within the projects I have been more focused on what is out there instead of the things I know and am more comfortable with. Meaning that I have been more focused on researching possibilities instead of going with what I know.

Areas for improvement:

- While I have been focused on taking a more research-based approach, I still tend to be held back due to wanting to do what I know.

Action items :

- Explore the options on how to help improve project management.

5.3 Sprint 3

5.3.1 Retrospective

What went well:

- After struggles with people showing up to college, we had an open conversation about the situation. Which really helped the team to come closer together and get a better understanding of each perspective.

Areas for improvement:

- While some of the team members are active with committing and keeping an eye on open merge requests, the others seem to not be consistent. Especially when it comes to the merge requests, as sometimes it takes days to be reviewed or updated based upon reviews.

Action items:

- Improve pipeline for automated testing, for example, setup a server, so the pipeline runner does not depend on whether the runner on my laptop is up and running.

5.3.2 Reflection

What went well:

- Overall communicated well with the team, and tried to explain where I could, and when necessary.

Areas for improvement:

- As I sometimes have the tendency take the lead too much, I could step down sometimes to ensure other feel the space to discuss their point of view. I do try to ask how they see it, etc., but I forget occasionally.

Action items :

- Focus more on what can be achieved and is feasible, and less on what could be achieved if there was more time.

5.4 Sprint 4

5.4.1 Retrospective

What went well:

- Been more active on meeting at campus, which boosted the productivity and overall way of working. Especially as the different parts had to start coming together.

Areas for improvement:

- When problems arise, don't get stuck on it too long. Find a different approach as too much time gets lost by trying to fix it.

Action items:

- Last demo demonstrate the whole process and little technical detail.

5.4.2 Reflection

What went well:

- Been able to be more at campus, and been more aware of what is going on within the team.

Areas for improvement:

- While wanting to help everyone when they need it, occasionally I should prioritize my own tasks as I get behind on them by helping everyone.

Action items :

- Give one last boost to complete the projects.

5.5 Sprint 5

6 Evaluation and reflection on the whole semester

For 'wrapping up' the semester. Add an evaluation and a reflection of your whole semester. Your evaluation describes what went good and bad during your process and how you dealt with that. Your reflection describes how you have grown as a person, and what you will take with you in your further professional career (e.g. towards the graduation internship).

7 Feedpulse

Checkpoint 5 22-11-2023



Overall well done. Involved the other team mates more in the process and their view points.



got everything working with the database and communication and was really helpful and helped other a lot when they were stuck



Laat goed weten wat ons te verwachten staat en hoe iedereen's deel eruit moet zien. Bespreekt tijdig design choices en helpt bij het opzetten van docker zodat ieder aan de slag kan. Good teammate

Checkpoint 4 22-11-2023



Generally speaking, I believe that I have progressed well and continue to take all the feedback in account. As I sometimes have the tendency take the lead too much, I could step down sometimes to ensure other feel the space too discuss their pov. I do try to ask how they see it, etc, but sometimes I forget.



Makes sure that everyone understands the whole meaning behind something. Helps in communicating within and outside the group. Likes to help others and give examples



Goed dat je dit sprint meer op school bent aanwezig. Probeer in het weekend het aantal berichten in Times te beperken.



gives good constructive feedback on peer review, sometimes went out of the scope of the user story with the feedback for the merge

Checkpoint 3 01-11-2023



Overall happy with my performance. I have been able to perform more tasks and be more available as illness is became more manageable. Tried to take the feedback from previous sprint and show progress more often by sharing updates , text based and image.



Came with the idea to do a weekly meeting to share each others progress and foundings, since then it is really good to follow whats going on. No minus points this sprint.



heeft database goed werkend en laat goed zien weten waar hij mee bezig is



Jij geeft duidelijk aan waar je aan werkt. Organiseert goed

Checkpoint 2 01-11-2023



Overall happy with my performance. I have been able to perform more tasks and be more available as illness is became more manageable. Tried to take the feedback from previous sprint and show progress more often by sharing updates , text based and image.



No feedback has been added.



Git netjes opgezet, en goed aan de project documenten gewerkt. Mag iets meer laten zien waar jij mee bezig bent.

Checkpoint 1 10-10-2023



Regardless of illness, still trying to do the things I am capable of doing. Should be more vocal towards the coach regarding to the team situation as there some struggles, which are not being vocalized outside direct team members.



setup the git and worked on communication but went sick and was mia



Git netjes opgezet, en goed aan de project documenten gewerkt. Mag iets meer laten zien waar jij mee bezig bent.



Great communication and setting up the GIT / work flow within the group. Only thing I can say is, get well soon. Maybe try to communicate slightly more with Gharib, he had to guess what database you went with (for his part in the project). Eventhough he couldve asked